

Session 5 Homework Solutions

APSTA-GE 2006: Applied Statistics for Social Science Research

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This document provides worked solutions to the [Lecture 5 homework](#).

1 Binomial PMF and CDF

Task: Let $X \sim \text{Binomial}(20, 0.35)$. Compute $\mathbb{P}(X = 7)$, $\mathbb{P}(X \leq 7)$, and $\mathbb{P}(5 \leq X \leq 10)$; verify each result with `dbinom()` or `pbinom()`, then plot the PMF and CDF for $x = 0, \dots, 20$.

For a binomial random variable with $N = 20$ and $\theta = 0.35$, the PMF is

$$\mathbb{P}(X = x) = \binom{20}{x} (0.35)^x (0.65)^{20-x}, \quad x = 0, 1, \dots, 20.$$

Therefore,

$$\mathbb{P}(X = 7) = \binom{20}{7} (0.35)^7 (0.65)^{13}.$$

The two interval probabilities can be written as

$$\begin{aligned} \mathbb{P}(X \leq 7) &= \sum_{x=0}^7 \binom{20}{x} (0.35)^x (0.65)^{20-x}, \\ \mathbb{P}(5 \leq X \leq 10) &= \mathbb{P}(X \leq 10) - \mathbb{P}(X \leq 4). \end{aligned}$$

```

N <- 20
theta <- 0.35

p_equal_7 <- dbinom(7, size = N, prob = theta)
p_at_most_7 <- pbinom(7, size = N, prob = theta)
p_5_through_10 <-
  pbinom(10, size = N, prob = theta) -
  pbinom(4, size = N, prob = theta)

binomial_answers <- data.frame(
  quantity = c("P(X = 7)", "P(X <= 7)", "P(5 <= X <= 10)"),
  probability = c(p_equal_7, p_at_most_7, p_5_through_10)
)

knitr::kable(
  binomial_answers,
  digits = 5,
  booktabs = TRUE,
  caption = "Binomial probability calculations"
)

```

Table 1: Binomial probability calculations

quantity	probability
P(X = 7)	0.18440
P(X <= 7)	0.60103
P(5 <= X <= 10)	0.82864

Thus,

$$\mathbb{P}(X = 7) = 0.18440, \quad \mathbb{P}(X \leq 7) = 0.60103, \quad \mathbb{P}(5 \leq X \leq 10) = 0.82864.$$

The PMF gives each point probability, while the CDF accumulates those probabilities from left to right.

```

binomial_grid <- data.frame(
  x = 0:N,
  pmf = dbinom(0:N, size = N, prob = theta),
  cdf = pbinom(0:N, size = N, prob = theta)
)

pmf_plot <- ggplot(binomial_grid, aes(x, pmf)) +
  geom_col(width = 0.7, fill = "#3564d4") +
  scale_x_continuous(breaks = seq(0, N, by = 2)) +
  labs(
    x = "x",
    y = "P(X = x)",
    title = "Probability mass function"
  )

```

```

)

cdf_plot <- ggplot(binomial_grid, aes(x, cdf)) +
  geom_step(direction = "hv", linewidth = 0.6, color = "#d95f02") +
  geom_point(size = 1.3, color = "#d95f02") +
  scale_x_continuous(breaks = seq(0, N, by = 2)) +
  scale_y_continuous(limits = c(0, 1)) +
  labs(
    x = "x",
    y = "P(X <= x)",
    title = "Cumulative distribution function"
  )

pmf_plot + cdf_plot

```

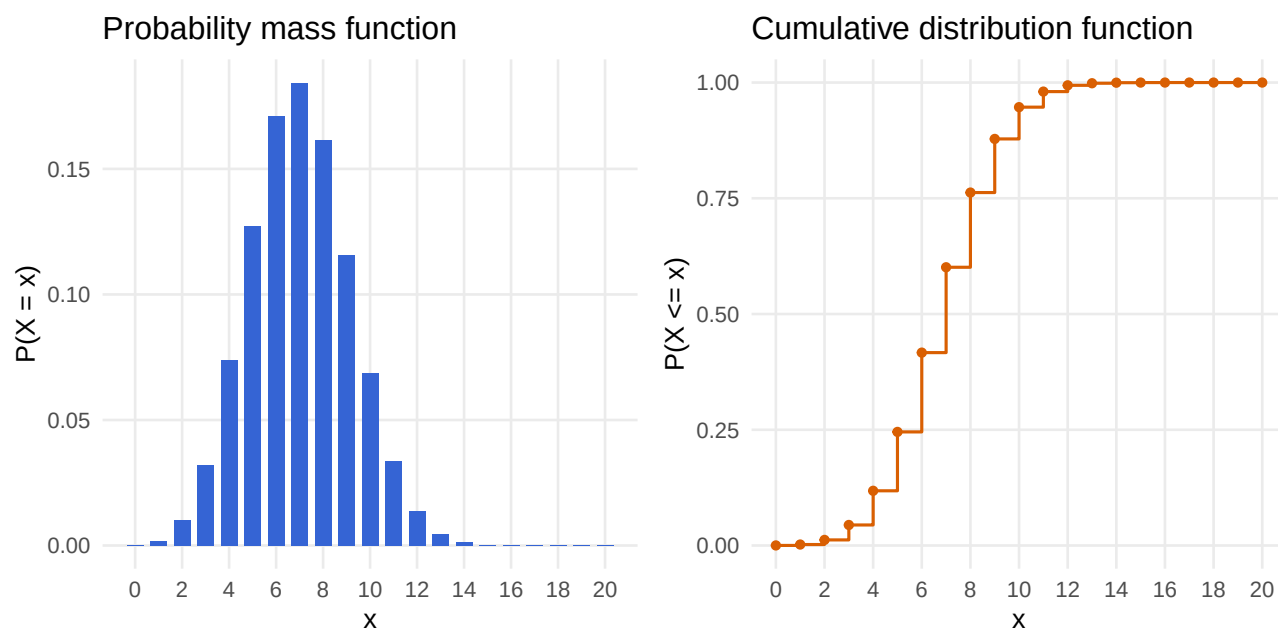


Figure 1: PMF and CDF of a Binomial(20, 0.35) random variable.

2 First-success simulation

Task: Let T be the trial on which the first success occurs when $\theta = 0.2$. Simulate 50,000 values with `rgeom() + 1`; for $t = 1, \dots, 10$, compare the empirical probabilities with $\mathbb{P}(T = t) = \theta(1 - \theta)^{t-1}$. Compare the simulated mean and variance with $1/\theta$ and $(1 - \theta)/\theta^2$.

R's `rgeom()` counts the number of failures before the first success. Adding 1 changes that output into the trial number of the first success.

For $T = 1, 2, \dots$,

$$\mathbb{P}(T = t) = \theta(1 - \theta)^{t-1}, \quad \mathbb{E}(T) = \frac{1}{\theta}, \quad \mathbb{V}(T) = \frac{1 - \theta}{\theta^2}.$$

With $\theta = 0.2$, the theoretical mean and variance are

$$\mathbb{E}(T) = 5, \quad \mathbb{V}(T) = 20.$$

```
set.seed(502)

geometric_theta <- 0.2
geometric_reps <- 50000
first_success <- rgeom(geometric_reps, prob = geometric_theta) + 1

estimate_empirical_probability <- function(value, simulated_values) {
  number_of_matches <- sum(simulated_values == value)
  number_of_simulations <- length(simulated_values)

  number_of_matches / number_of_simulations
}

geometric_comparison <- data.frame(t = 1:10) |>
  mutate(
    empirical = purrr::map_dbl(
      t,
      estimate_empirical_probability,
      simulated_values = first_success
    ),
    theoretical = geometric_theta * (1 - geometric_theta)^(t - 1),
    difference = empirical - theoretical
  )

knitr::kable(
  geometric_comparison,
  digits = 5,
  booktabs = TRUE,
  caption = "Empirical and theoretical first-success probabilities"
)
```

Table 2: Empirical and theoretical first-success probabilities

t	empirical	theoretical	difference
1	0.19802	0.20000	-0.00198
2	0.15734	0.16000	-0.00266
3	0.12846	0.12800	0.00046
4	0.10434	0.10240	0.00194
5	0.08062	0.08192	-0.00130
6	0.06732	0.06554	0.00178
7	0.05222	0.05243	-0.00021

t	empirical	theoretical	difference
8	0.04186	0.04194	-0.00008
9	0.03288	0.03355	-0.00067
10	0.02668	0.02684	-0.00016

The pointwise simulation error is small and changes sign across support points, as expected from Monte Carlo variation.

```
geometric_comparison |>
  select(t, empirical, theoretical) |>
  pivot_longer(
    cols = c(empirical, theoretical),
    names_to = "source",
    values_to = "probability"
  ) |>
  ggplot(aes(t, probability, color = source, shape = source)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 1:10) +
  scale_color_manual(
    values = c(empirical = "#3564d4", theoretical = "#d95f02"),
    labels = c(empirical = "simulation", theoretical = "theory")
  ) +
  scale_shape_manual(
    values = c(empirical = 16, theoretical = 1),
    labels = c(empirical = "simulation", theoretical = "theory")
  ) +
  labs(
    x = "trial of first success, t",
    y = "probability",
    color = NULL,
    shape = NULL
  )
```

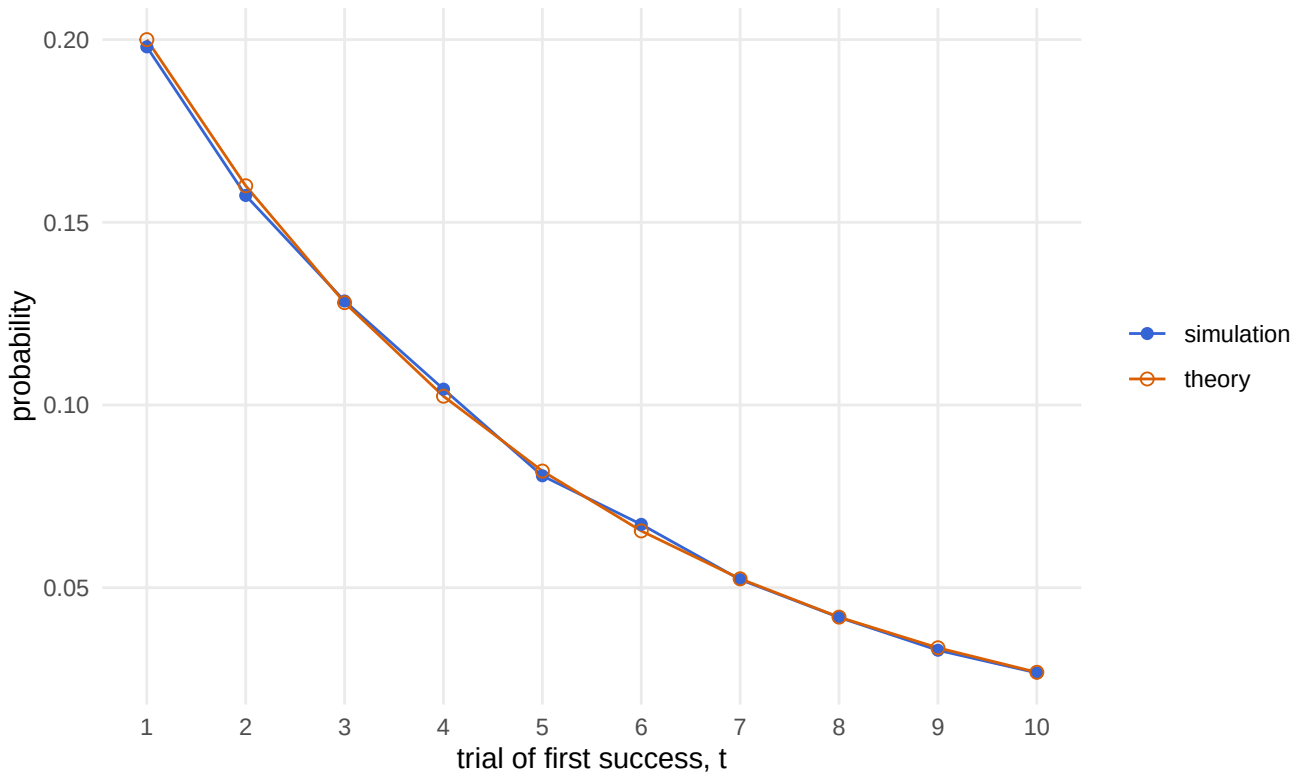


Figure 2: Simulated and theoretical probabilities for the first-success trial.

```
geometric_moments <- data.frame(
  source = c("simulation", "theory"),
  mean = c(mean(first_success), 1 / geometric_theta),
  variance = c(
    var(first_success),
    (1 - geometric_theta) / geometric_theta^2
  )
)

knitr::kable(
  geometric_moments,
  digits = 4,
  booktabs = TRUE,
  caption = "First-success mean and variance"
)
```

Table 3: First-success mean and variance

source	mean	variance
simulation	5.0369	20.1046
theory	5.0000	20.0000

The simulated mean is 5.037 and the simulated variance is 20.105, both close to 5 and 20.

3 Rare events: binomial and Poisson

Task: Simulate 100,000 values from $\text{Binomial}(1000, 0.003)$ and compare the empirical PMF for $0, \dots, 10$ with a Poisson PMF having $\lambda = 3$. Compare the two means, variances, and probabilities of at least 6 events.

Here

$$N = 1000, \quad \theta = 0.003, \quad N\theta = 3.$$

The binomial mean and variance are

$$\mathbb{E}(X) = N\theta = 3, \quad \mathbb{V}(X) = N\theta(1 - \theta) = 2.991.$$

A $\text{Poisson}(3)$ random variable has mean and variance equal to 3, so the two distributions should be close.

```
set.seed(503)

rare_reps <- 100000
rare_size <- 1000
rare_prob <- 0.003
poisson_lambda <- rare_size * rare_prob

rare_counts <- rbinom(
  rare_reps,
  size = rare_size,
  prob = rare_prob
)

# Reuse the empirical-probability function from Problem 2
rare_pmf <- data.frame(k = 0:10) |>
  mutate(
    empirical_binomial = purrr::map_dbl(
      k,
      estimate_empirical_probability,
      simulated_values = rare_counts
    ),
    exact_binomial = dbinom(k, size = rare_size, prob = rare_prob),
    poisson = dpois(k, lambda = poisson_lambda)
  )

knitr::kable(
  rare_pmf,
  digits = 5,
  booktabs = TRUE,
  caption = "Rare-event PMF comparison"
)
```

Table 4: Rare-event PMF comparison

k	empirical_binomial	exact_binomial	poisson
0	0.04978	0.04956	0.04979
1	0.14872	0.14914	0.14936
2	0.22280	0.22415	0.22404
3	0.22404	0.22438	0.22404
4	0.16857	0.16828	0.16803
5	0.10172	0.10087	0.10082
6	0.05027	0.05033	0.05041
7	0.02216	0.02151	0.02160
8	0.00835	0.00803	0.00810
9	0.00254	0.00266	0.00270
10	0.00083	0.00079	0.00081

```

rare_pmf |>
  pivot_longer(
    cols = -k,
    names_to = "source",
    values_to = "probability"
  ) |>
  ggplot(aes(k, probability, color = source, shape = source)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.8) +
  scale_x_continuous(breaks = 0:10) +
  scale_color_manual(
    values = c(
      empirical_binomial = "#222222",
      exact_binomial = "#3564d4",
      poisson = "#d95f02"
    ),
    labels = c(
      empirical_binomial = "simulated binomial",
      exact_binomial = "exact binomial",
      poisson = "Poisson comparison"
    )
  ) +
  scale_shape_manual(
    values = c(
      empirical_binomial = 16,
      exact_binomial = 1,
      poisson = 2
    ),
    labels = c(
      empirical_binomial = "simulated binomial",
      exact_binomial = "exact binomial",
      poisson = "Poisson comparison"
    )
  ) +

```

```

labs(
  x = "event count",
  y = "probability",
  color = NULL,
  shape = NULL
)

```

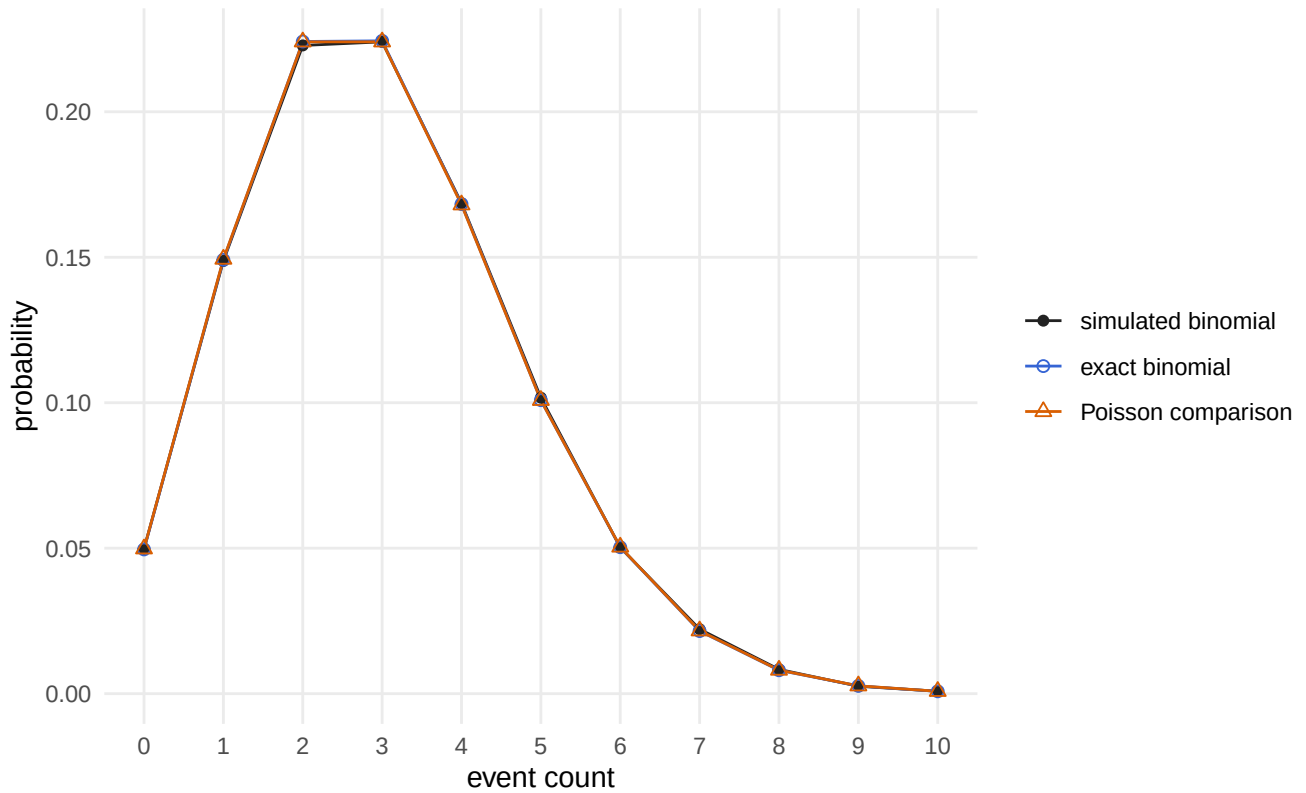


Figure 3: The empirical binomial PMF and its Poisson comparison.

For the upper-tail probability,

$$\mathbb{P}(X \geq 6) = 1 - \mathbb{P}(X \leq 5).$$

```

rare_summary <- data.frame(
  source = c(
    "simulated binomial",
    "exact binomial",
    "Poisson comparison"
  ),
  mean = c(
    mean(rare_counts),
    rare_size * rare_prob,
    poisson_lambda
  ),

```

```

variance = c(
  var(rare_counts),
  rare_size * rare_prob * (1 - rare_prob),
  poisson_lambda
),
probability_at_least_6 = c(
  mean(rare_counts >= 6),
  pbinom(5, size = rare_size, prob = rare_prob, lower.tail = FALSE),
  ppois(5, lambda = poisson_lambda, lower.tail = FALSE)
)
)

knitr::kable(
  rare_summary,
  digits = 5,
  booktabs = TRUE,
  caption = "Rare-event summary comparison"
)

```

Table 5: Rare-event summary comparison

source	mean	variance	probability_at_least_6
simulated binomial	3.00654	3.00483	0.08437
exact binomial	3.00000	2.99100	0.08361
Poisson comparison	3.00000	3.00000	0.08392

The exact binomial and Poisson upper-tail probabilities differ by only 0.00030. The approximation works well because there are many trials and a small success probability.

4 Checking Poisson equidispersion

Task: Repeat 1,000 times: simulate 200 counts from $\text{Poisson}(4)$ and record their sample mean and variance. Plot variance against mean. Then repeat after drawing each count's rate independently from $\{1, 7\}$ with equal probability. Compare the plots and explain why the heterogeneous counts are overdispersed relative to a single Poisson model.

For the homogeneous model, every count has the same rate:

$$X_i \sim \text{Poisson}(4).$$

Its population mean and variance are both 4. We therefore expect each dataset's sample mean and variance to fluctuate around the diagonal point (4, 4).

For the heterogeneous model, each observation first receives rate 1 or rate 7 with equal probability. The average rate remains

$$\frac{1+7}{2} = 4.$$

However, the rates themselves vary. Their variance is

$$\frac{(1-4)^2 + (7-4)^2}{2} = 9.$$

That between-rate variation adds to the average within-rate Poisson variance of 4, giving an overall count variance of

$$4 + 9 = 13.$$

Thus, both models have mean 4, but only the homogeneous model has variance 4.

```
set.seed(504)

count_reps <- 1000
counts_per_dataset <- 200

simulate_count_summaries <- function(repetitions, draw_counts) {
  results <- replicate(repetitions, {
    counts <- draw_counts()
    c(
      sample_mean = mean(counts),
      sample_variance = var(counts)
    )
  })

  as.data.frame(t(results))
}

homogeneous <- simulate_count_summaries(
  count_reps,
  function() rpois(counts_per_dataset, lambda = 4)
) |>
  mutate(model = "single rate: lambda = 4")

heterogeneous <- simulate_count_summaries(
  count_reps,
  function() {
    rates <- sample(
      c(1, 7),
      size = counts_per_dataset,
      replace = TRUE
    )
    rpois(counts_per_dataset, lambda = rates)
  }
) |>
```

```

mutate(model = "mixed rates: 1 or 7")

poisson_checks <- bind_rows(homogeneous, heterogeneous)

poisson_summary <- poisson_checks |>
  group_by(model) |>
  summarise(
    average_sample_mean = mean(sample_mean),
    average_sample_variance = mean(sample_variance),
    sd_of_sample_means = sd(sample_mean),
    .groups = "drop"
  )

poisson_summary_display <- poisson_summary |>
  transmute(
    Model = model,
    `Average mean` = average_sample_mean,
    `Average variance` = average_sample_variance,
    `SD of means` = sd_of_sample_means
  )

knitr::kable(
  poisson_summary_display,
  digits = 3,
  booktabs = TRUE,
  caption = "Repeated-simulation summaries"
)

```

Table 6: Repeated-simulation summaries

Model	Average mean	Average variance	SD of means
mixed rates: 1 or 7	4.012	13.038	0.257
single rate: $\lambda = 4$	3.998	3.991	0.138

```

poisson_checks |>
  ggplot(aes(sample_mean, sample_variance)) +
  geom_abline(
    slope = 1,
    intercept = 0,
    color = "#d95f02",
    linewidth = 0.6,
    linetype = "dashed"
  ) +
  geom_point(color = "#3564d4", alpha = 0.25, size = 0.9) +
  facet_wrap(~ model) +
  scale_x_continuous(
    limits = c(0, 18),
    breaks = seq(0, 18, by = 3),
  )

```

```

    expand = expansion(mult = 0.02)
) +
scale_y_continuous(
  limits = c(0, 18),
  breaks = seq(0, 18, by = 3),
  expand = expansion(mult = 0.02)
) +
coord_fixed(ratio = 1) +
labs(
  x = "sample mean",
  y = "sample variance"
)

```

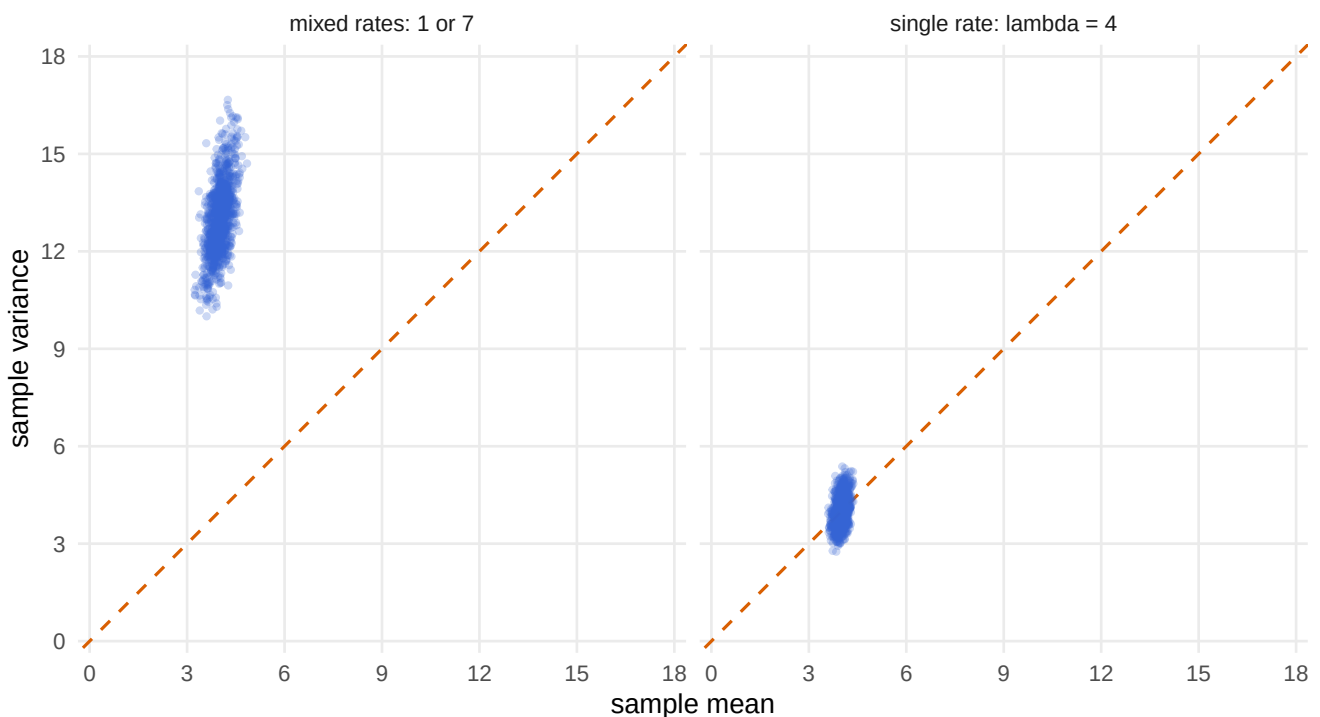


Figure 4: Sample variance versus sample mean for homogeneous and heterogeneous Poisson counts.

Both axes now run from 0 to 18 with a 1:1 aspect ratio, so equal numerical changes occupy equal visual distances. The nearly vertical clouds show that the sample means vary within a narrow range. The homogeneous simulations cluster near mean 4 and variance 4, close to the dashed equidispersion line. The heterogeneous simulations retain mean 4 but cluster near variance 13. A single Poisson model with rate 4 cannot reproduce that extra variation, so the heterogeneous counts are overdispersed relative to it.

5 Main conclusions

1. `dbinom()` evaluates a binomial point probability, while `pbinom()` accumulates probabilities through a cutoff.

2. A geometric simulation using `rgeom() + 1` agrees with the first-success PMF, mean, and variance.
3. A Poisson PMF closely approximates a binomial PMF when the number of trials is large, the success probability is small, and $N\theta$ is held fixed.
4. Poisson equidispersion can fail when observations have different rates, even when their average rate is unchanged.